

Docker Usage Guide

This guide covers Docker deployment options for the Trivy Security Dashboard.

Quick Start with Docker Hub

Pull and Run

```
# Pull the latest stable image
docker pull yourusername/trivy-security-dashboard:latest

# Run with McCamish Nexus configuration
docker run -d \
  --name trivy-dashboard \
  -p 5000:5000 \
  -e NEXUS_URL=http://swdlvapp682.mccamish.com:8081 \
  -e NEXUS_USERNAME=your-username \
  -e NEXUS_PASSWORD=your-password \
  -e NEXUS_REPOSITORY=mccamish_sbom \
  -e NEXUS_GROUP_ID=com.mccamish \
  -e NEXUS_ARTIFACT_SUFFIX=.sbom \
  yourusername/trivy-security-dashboard:latest
```

Access Dashboard

- Dashboard: <http://localhost:5000>
- Health Check: <http://localhost:5000/api/health>
- Debug Info: <http://localhost:5000/debug/nexus>

Environment Configuration

Required Variables

NEXUS_URL=http://your-nexus.company.com:8081	# Nexus Repository URL
NEXUS_USERNAME=your-username	# Nexus authentication
username	
NEXUS_PASSWORD=your-password	# Nexus authentication
password	
NEXUS_REPOSITORY=your-sbom-repository	# Repository containing SBOM
files	

Generic Pattern Variables (customize for your setup)

```
# For Jenkins upload pattern: groupId/artifactId/version/filename
NEXUS_GROUP_ID=com.mccamish           # Maven groupId
NEXUS_ARTIFACT_SUFFIX=.sbom           # Suffix for artifactId
NEXUS_VERSION_PREFIX=1.0.0-          # Version prefix before
timestamp
NEXUS_ASSET_EXTENSION=json           # File extension for SBOM files
```

Optional Configuration

```
# Application Settings
FLASK_DEBUG=false                     # Enable debug mode
LOG_LEVEL=INFO                       # Logging level (DEBUG, INFO,
WARNING, ERROR)
CACHE_TTL=300                        # Cache time-to-live in seconds
DATA_REFRESH_INTERVAL=600            # Background refresh interval in
seconds

# Performance Settings
NEXUS_TIMEOUT=30                     # Nexus API timeout in seconds
MAX_SCANS_PER_PROJECT=50             # Maximum scans to keep per
project
```

Docker Compose Deployment

Complete Stack

```
version: '3.8'

services:
  trivy-dashboard:
    image: yourusername/trivy-security-dashboard:latest
    container_name: trivy-dashboard
    ports:
      - "5000:5000"
    environment:
      # Nexus Configuration
      - NEXUS_URL=http://nexus:8081
      - NEXUS_USERNAME=admin
      - NEXUS_PASSWORD=admin123
      - NEXUS_REPOSITORY=trivy-reports

      # Generic Pattern (customize for your setup)
      - NEXUS_GROUP_ID=com.example
      - NEXUS_ARTIFACT_SUFFIX=.sbom
      - NEXUS_VERSION_PREFIX=1.0.0-
      - NEXUS_ASSET_EXTENSION=json
```

```

    # Application Settings
    - FLASK_ENV=production
    - LOG_LEVEL=INFO
    - CACHE_TTL=300
    - DATA_REFRESH_INTERVAL=600
volumes:
    - ./logs:/app/logs
restart: unless-stopped
depends_on:
    - nexus
networks:
    - trivy-network

# Optional: Include Nexus if you don't have one
nexus:
    image: sonatype/nexus3:latest
    container_name: nexus-repository
    ports:
        - "8081:8081"
    volumes:
        - nexus-data:/nexus-data
    environment:
        - INSTALL4J_ADD_VM_PARAMS=-Xms1g -Xmx1g
    restart: unless-stopped
    networks:
        - trivy-network

networks:
    trivy-network:
        driver: bridge

volumes:
    nexus-data:
        driver: local

```

Start Services

```

# Start all services
docker-compose up -d

# View logs
docker-compose logs -f trivy-dashboard

# Stop services
docker-compose down

```

Troubleshooting

Debug Connection Issues

```
# Check debug endpoint
curl http://localhost:5000/debug/nexus

# View container logs
docker logs trivy-dashboard

# Check health status
curl http://localhost:5000/api/health
```

Common Issues

No SBOM Files Found

```
# Verify Nexus configuration
docker exec trivy-dashboard env | grep NEXUS

# Check if repository exists and contains files
curl -u username:password "http://nexus:8081/service/rest/v1/repositories"
```

Connection Refused

```
# Check if Nexus is accessible from container
docker exec trivy-dashboard curl -I http://your-nexus:8081

# Verify network connectivity
docker network ls
docker network inspect <network_name>
```

Permission Denied

```
# Check authentication
curl -u username:password "http://nexus:8081/service/rest/v1/status"

# Verify credentials in environment
docker exec trivy-dashboard env | grep -E "(NEXUS_USERNAME|NEXUS_PASSWORD)"
```

Production Deployment

With Reverse Proxy (Nginx)

```
server {
    listen 80;
    server_name trivy-dashboard.company.com;

    location / {
        proxy_pass http://localhost:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

With SSL/TLS

```
# Generate SSL certificate (Let's Encrypt example)
certbot --nginx -d trivy-dashboard.company.com

# Update Docker run command
docker run -d \
    --name trivy-dashboard \
    -p 127.0.0.1:5000:5000 \
    --env-file .env \
    --restart unless-stopped \
    yourusername/trivy-security-dashboard:latest
```

Resource Limits

```
# Run with resource constraints
docker run -d \
    --name trivy-dashboard \
    -p 5000:5000 \
    --memory=512m \
    --cpus=1.0 \
    --env-file .env \
    --restart unless-stopped \
    yourusername/trivy-security-dashboard:latest
```

Monitoring

Health Checks

```
# Built-in health check
curl http://localhost:5000/api/health
```

```
# Docker health status
docker inspect --format='{{.State.Health.Status}}' trivy-dashboard
```

Metrics Collection

```
# Container stats
docker stats trivy-dashboard

# Application logs
docker logs -f trivy-dashboard --tail 100
```

Alerts

Set up monitoring for:

- Container health status
- Nexus connectivity
- Dashboard response time
- Error rates in logs

Updates

Update to Latest Version

```
# Stop current container
docker stop trivy-dashboard
docker rm trivy-dashboard

# Pull latest image
docker pull yourusername/trivy-security-dashboard:latest

# Start with same configuration
docker run -d \
  --name trivy-dashboard \
  -p 5000:5000 \
  --env-file .env \
  --restart unless-stopped \
  yourusername/trivy-security-dashboard:latest
```

Using Docker Compose

```
# Update and restart
docker-compose pull
docker-compose up -d
```

This Docker setup provides a robust, scalable deployment option for the Trivy Security Dashboard with full configuration flexibility.